Effects of concurrent use of alcohol and cocaine

Ed J. M. Pennings, Arthur P. Leccese & Frederik A. de Wolff

Toxicology Laboratory, Leiden University Medical Center, Leiden, The Netherlands

Correspondence to:
Dr E. J. M. Pennings
Toxicology Laboratory
L1-P, Leiden University Medical Center
PO Box 9600
2300 RC Leiden
The Netherlands

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ABSTRACT

The combination of alcohol and cocaine is popular among drug users, perhaps because of more intense feelings of 'high' beyond that perceived with either drug alone, less intense feelings of alcohol-induced inebriation and tempering of discomfort when coming down from a cocaine 'high'. A review is presented of the medical literature on psychological and somatic effects and consequences of combined use of alcohol and cocaine in man. The search was carried out with Medline, the Science Citation Index/Web of Science and Toxline. Exclusion and inclusion criteria for this search are identified. There is generally no evidence that the combination of the two drugs does more than enhance additively the already strong tendency of each drug to induce a variety of physical and psychological disorders. A few exceptions must be noted. Cocaine consistently antagonizes the learning deficits, psychomotor performance deficits and driving deficits induced by alcohol. The combination of alcohol and cocaine tends to have greater-than-additive effects on heart rate, concomitant with up to 30% increased blood cocaine levels. Both prospective and retrospective data further reveal that co-use leads to the formation of cocaethylene, which may potentiate the cardiotoxic effects of cocaine or alcohol alone. More importantly, retrospective data suggest that the combination can potentiate the tendency towards violent thoughts and threats, which may lead to an increase of violent behaviours.

KEYWORDS Alcohol-related disorders, cocaine, cocaine-related disorders, drug interactions, ethanol, human.

INTRODUCTION

This survey presents the results of a review of the literature about the psychological and somatic effects of combined use of alcohol and cocaine. We describe the evidence regarding whether alcohol and cocaine can enhance or suppress the effects of each other when used in combination, and whether such interactions are additive or potentiating. All identified reviews, prospective studies and retrospective studies on combined use and interactions of alcohol and cocaine have been examined. Studies that involve comparison, rather than combined use of alcohol and cocaine, are excluded from this review.

In order to quell any potential objections based on species differences this review focuses only on research involving humans.

METHODS

Search terms were chosen to extract studies describing the effects and consequences of combined use of alcohol and cocaine. Alcohol terms, drug terms and drug interaction terms were combined with Boolean operators to extract retrospective and prospective studies. Alcohol search terms included: alcohol, ethanol, alcohol drinking, alcohol-related disorder, alcohol amnestic disorder, fetal alcohol syndrome, cardiomyopathy—alcoholic, hepatitis—alcoholic, fatty liver—alcoholic, liver cirrhosis—alcoholic, pancreatitis—alcoholic, psychosis—alcoholic. Cocaine terms included: cocaine, crack cocaine. Drug interaction terms included: drug interactions, drug synergism, drug toxicity, substance-related disorders, psychoses—substance-induced.

The databases searched included Medline (1963–present), Science Citation Index/Web of Science (1988–present) and Toxline (1980–present). The search was conducted in May–June 1999 and repeated in October 2001 before revision of the manuscript.

Inclusion and exclusion

All retrospective studies that enabled identification of subjects as users of alcohol plus cocaine, are included in this review. Prospective studies are included on the basis of satisfaction of criteria regarding the use of placebos, blind conditions of evaluation, cross-over design and appropriate presentation of data and statistics (Table 1). Excluded are papers that compared only the effects of alcohol with those of cocaine, studies not involving humans, studies referring to 'polydrug' abuse, and studies not in the English language. Two studies with a between-subjects design are included, despite design flaws brought about by ethical concerns regarding invasive technologies. One involved heart catherization for a medical reason (Pirwitz *et al*. 1995) and the other involved injection of trace amounts of radiolabelled substances for nuclear imaging (Volkow & Fowler 1994). Prospective experiments always involved either the insufflation or the intravenous injection, and never the smoking, of cocaine.

Only a few of the studies satisfied Berenbaum's strict criterion of at least two active doses of each drug under consideration to allow any conclusions on interactions (Berenbaum 1977; Berenbaum 1989). Rather than rejecting the papers failing to satisfy this criterion, the term 'potentiation' will be used in deference to the stricter definition of Berenbaum's 'synergism'. Thus, interactions may be merely 'additive' or 'subtractive' (where the net effect is the sum of the expected effect of both substances acting alone), or they may be 'potentiating' or 'inhibitive' (where the net effect is greater, or less, than just the sum of each expected effect alone).

RESULTS

While most reviews of cocaine–alcohol interactions are general (Dial 1992; Gorelick 1992; Ritz, Kuhar & George 1992; Sands & Ciraulo 1992), some deal specifically with methodological (Jones 1986) or epidemiological issues (Smith 1986; Grant & Harford 1990), one with aggressive behaviour (Denison, Paredes & Booth 1997), some with metabolism and cocaethylene (Hoyumpa 1984; Jatlow *et al*. 1991; Landry 1992; Jatlow *et al*. 1996; Cami *et al*. 1998) and some with immunological (Donahoe & Klein 1996) or cardiovascular (Bunn & Giannini 1992) consequences. The main results of the analysis of the prospective and retrospective studies have been condensed into Tables 2–4.

Prospective research

Behavioural

Alcohol-induced behavioural deficits in the laboratory have been antagonized consistently by cocaine. For example, alcohol-induced deficits in the digit-symbol substitution test were antagonized by cocaine (Foltin *et al*. 1993; Higgins *et al*. 1993). Similarly, cocaine antagonized alcohol-induced sedation (Foltin *et al*. 1993) and deficits in both learning and psychomotor performance (Higgins *et al*. 1992).

Heart rate

Both cocaine and alcohol may increase heart rate, and when cocaine is given at the same time or after dosing with alcohol most studies show a trend towards greater-than-additive effects on heart rate but do not support the notion that the effects are synergistic (Foltin & Fischman 1988; Perez-Reyes & Jeffcoat 1992; Farré *et al*. 1993; Higgins *et al*. 1993; McCance-Katz *et al*. 1993; Farré *et al*. 1997; McCance-Katz, Kosten & Jatlow 1998). Prospective studies with at least two active doses of each drug are needed to clarify the issue of synergism. In another study (limited by a between-subjects design), the cocaine–alcohol combination caused an increase in the rate-pressure product (a determinant of myocardial oxygen demand) with a concomitant increase in epicardial coronary arterial diameter (Pirwitz *et al*. 1995). As the effects of cocaine alone can induce cardiovascular complications, the effects of co-use of cocaine and alcohol may put the combined user at serious clinical risk for cardiotoxicity (Bunn & Giannini 1992).

Perez-Reyes (1994) went beyond the usual paradigm to explicitly investigate drug dosing order effects. When cocaine is given 30 minutes before alcohol the increases in heart rate are additive (Perez-Reyes 1994). This contrasts with their earlier findings, that alcohol given (as is more common) before cocaine did indeed induce the commonly found greater-than-additive increases in heart rate (Perez-Reyes & Jeffcoat 1992). The higher blood levels of cocaine seen only when alcohol is given

Table 1 Summary of design and results from prospective studies of human concurrent use of alcohol and cocaine.

Authors	*Subjects*	*Placebo/blind cross-over*	*Sequence*	*Alcohol dose*	*Cocaine dose*[a]	*Interaction and comments*
Farré *et al.* (1993)	9	Yes/double cross-over	Alcohol 1st by 0 min	1 g/kg	100 mg	Non-significantly reduced drunk, more 'high'; greater-than-additive effects on heart rate; increased blood cocaine levels; production of cocaethylene
Farré *et al.* (1997)	8	Yes/double cross-over	Alcohol 1st by 0 min	0.8 g/kg	100 mg	Non-significantly reduced drunk, more 'high'; greater-than-additive effects on heart rate; increased blood cocaine levels; production of cocaethylene
Foltin & Fischman (1988)	9	Yes/single cross-over	Alcohol 1st by 25 min	19.4, 38.7 or 58.1 g	48 or 96 mg	Greater-than-additive effects on heart rate with smaller doses, additive effects with higher doses
Foltin *et al.* (1993)	12	Yes/single cross-over	Alcohol 1st by 25 min	19.4, 38.1 or 58.1 g	48 or 96 mg	Antagonism of alcohol sedation and alcohol-reduced digit-symbol-substitution-task performance
Higgins *et al.* (1992)	8	Yes/double cross-over	Alcohol 1st by 45 min	0.5 or 1 g/kg	48 or 96 mg/70 kg	Antagonism of alcohol-induced deficits in learning and performance
Higgins *et al.* (1993)	8	Yes/double cross-over	alcohol 1st by 45 min	0.5 or 1 g/kg	48 or 96 mg/70 kg	Greater-than-additive increases on heart rate; antagonism of alcohol-induced deficits; no or variable effects on 'high'
Higgins *et al.* (1996)	11	Yes/double cross-over	Alcohol 1st by 30 min	0.5 or 1 g/kg	max. cumulative dose: 100 mg	Alcohol increases preference for cocaine over cash
Mannelli *et al.* (1993)	10	Yes/double cross-over	Alcohol 1st by 40 min	0.5, or 1 g/kg	0.6 mg/kg i.v.	Additive effect (intensity and length) on 'high' and EEG
McCance-Katz *et al.* (1993)	6	Yes/double cross-over	Cocaine 1st by 0 min	1 g/kg	2 mg/kg	Additive increases in heart rate, and length and intensity of 'high'; cocaethylene formed and plasma cocaine up 30% with alcohol
McCance-Katz *et al.* (1998)	8	Yes/double cross-over	See note[b]			Greater-than-additive increases in heart rate and 'any high'; cocaethylene formed and plasma cocaine up 15% with alcohol
Perez-Reyes & Jeffcoat (1992)	11	Yes/single cross-over	Alcohol 1st by 15 min	0.85 g/kg	1.25 or 1.9 mg/kg	No effect on alcohol inebriation ratings; additive increases in heart rate; increased cocaine 'high' and plasma cocaine levels; formation of cocaethylene
Perez-Reyes (1994)	6	Yes/single cross-over	Alcohol 2nd by 35 min	0.85 g/kg	1.25 or 1.9 mg/kg	No increase in plasma cocaine; no effect on heart rate or 'high'; test of order effects; alcohol given second
Pirwitz *et al.* (1995)	34	Yes/single within	Cocaine 1st by 0 min	5 ml 10% (0.4 g)/kg	2 mg/kg	Increases of (determinants of) heart oxygen demand with concomitant increases of epicardial coronary arterial diameter
Volkow & Fowler (1994)	7	No/no within	Alcohol 1st by 40 min	1 g/kg	[^{11}C] cocaine tracer i.v.	No effect on cocaine uptake or clearance from brain and heart

[a] A 4-mg or 5-mg dose of cocaine is used in most studies as the placebo. Except in two studies with intravenous (i.v.) administration, cocaine was taken intranasally.
[b] Four doses of intranasal cocaine (1 mg/kg): first dose of cocaine at 0 minutes followed immediately by alcohol (1 g/kg); second dose of cocaine at 30 minutes; third dose of cocaine at 60 minutes followed immediately by second alcohol (120 mg/kg); fourth dose of cocaine at 90 minutes.

Table 2 Summary of effects of combined use, supported by prospective studies: effects relative to alcohol alone.

Alcohol alone (dose-dependent effects)	*Alcohol and cocaine: effects relative to alcohol alone*
Increased blood alcohol levels	Slightly decreased by high-dose cocaine; formation of cocaethylene
Increased subjective feeling of inebriation	Slightly decreased
Slightly increased heart rate	Trend towards greater-than-additive increase
Impaired cognitive and motor function	Less impaired

Table 3 Summary of effects of combined use, supported by prospective studies: effects relative to cocaine alone.

Cocaine alone (dose-dependent effects)	*Alcohol and cocaine: effects relative to cocaine alone*
Increased blood cocaine levels	Up to 30% further increased; formation of cocaethylene
Increased subjective feeling of 'high'	Greater-than-additive increase
Significantly increased heart rate	Trend towards greater-than-additive increase
Variable effects on cognitive and motor function	Impaired

Table 4 Summary of effects of combined use, supported by retrospective studies.

Alcohol alone (dose-dependent effects)	*Alcohol and cocaine*
Increased risk of committing violent acts	Additive effects on behaviour and crime; increased risk of homocidal thoughts and plans
Increased risk of psychiatric disorder	No reports found
Increased risk of birth defects	More-than-additive risk (one report found)
Impaired breathing	No reports found

before or during cocaine absorption (see next paragraph) may explain this phenomenon.

Blood levels and toxicokinetics

When compared to cocaine alone, cocaine given at the same time as or after alcohol resulted in an up to 30% increase in blood levels of cocaine (Perez-Reyes & Jeffcoat 1992; Farré *et al*. 1993; McCance-Katz *et al*. 1993; Farré *et al*. 1997; McCance-Katz, Kosten & Jatlow 1998). When compared to alcohol alone, the combination either had no effect (Perez-Reyes & Jeffcoat 1992) or produced a slight, non-significant decrease (about 10%) of blood alcohol levels (Farré *et al*. 1993; Higgins *et al*. 1993; McCance-Katz *et al*. 1993; Farré *et al*. 1997). In these studies, cocaine was given after completion of alcohol intake. One possible explanation is the decreased absorption of alcohol secondary to vasoconstriction after cocaine. The formation of cocaethylene is too small to account for the lower blood alcohol levels.

As with heart rate, reversing the order of administration can eliminate this effect of the combination on blood cocaine concentration. Thus, when cocaine was given to subjects 30 minutes before alcohol, there was no increase in blood cocaine levels when compared to subjects given cocaine alone (Perez-Reyes 1994). The increase in cocaine levels when alcohol is given before or at the same time as cocaine may be due to the competitive inhibition of cocaine-metabolizing esterases by alcohol (Landry 1992; Jatlow *et al*. 1996; Cami *et al*. 1998).

Alcohol had no effect upon the uptake or clearance of cocaine from either the heart or brain of subjects given an infusion of tracer doses of [^{11}C] cocaine (Volkow & Fowler 1994). Unfortunately, no placebo was included, subjects and evaluators were not blind, responses to cocaine or alcohol alone were not determined and results from the tracer dose of [^{11}C] cocaine may not generalize to behaviourally active doses of cocaine.

Effect on perceived inebriation

In several studies, cocaine produced a slight decrease (Farré *et al*. 1993; Higgins *et al*. 1993; Farré *et al*. 1997), while in another study cocaine had no effect upon alcohol-induced inebriation (Perez-Reyes & Jeffcoat 1992). Furthermore, alcohol given before or simultaneous with cocaine led to a greater subjective perception of cocaine 'high' or 'global intoxication' (Perez-Reyes & Jeffcoat 1992; Farré *et al*. 1993; Mannelli *et al*. 1993; McCance-Katz *et al*. 1993; Farré *et al*. 1997; McCance-

Katz, Kosten & Jatlow 1998). The intensity and length of the increase in 'high' was confirmed in one study by a well-correlated combination-induced increase in intensity and duration of EEG indices of intoxication (Mannelli *et al*. 1993). Consistent with the hypothesis that simultaneous use leads to an at least additive 'high', alcohol use increased the preference for cocaine in subjects offered a postalcohol choice between cocaine and cash (Higgins, Roll & Bickel 1996).

Interestingly, when the order of administration was reversed so that cocaine was given first and alcohol second (and there was thus no interactive effect of concurrent use upon heart rate or blood levels of cocaine) there was no effect of the combination upon the feeling of being 'high' (Perez-Reyes 1994). This result might arise from the absence of alcohol during the first 30 minutes of cocaine intoxication, or instead from some sort of suppressive effect of later alcohol. This has been studied in an experiment with multiple dosings of each compound (McCance-Katz, Kosten & Jatlow 1998). Subjects experienced the usual 'alcohol first' pattern of effects, with greater-than-additive increases in heart rate, plasma cocaine and 'high'. These results support the first interpretation that the lack of impact from the combination where alcohol is given last arises from the absence of alcohol during the initial absorption, distribution and metabolism of cocaine, rather than from some suppressive effect of the final dose of pretesting alcohol.

Cocaethylene production

Experimental subjects produced cocaethylene only when they were given the combination, and never when they were given either alcohol or cocaine alone (Perez-Reyes & Jeffcoat 1992; Farré *et al*. 1993; McCance-Katz *et al*. 1993; Perez-Reyes 1994; Farré *et al*. 1997; McCance-Katz, Kosten & Jatlow 1998). Cocaethylene was also formed when alcohol was given 30 minutes after cocaine snorting (Perez-Reyes 1994).

Cocaethylene may play a role in the increased heart rate and 'high' perceived by those who simultaneously use alcohol and cocaine. It has similar behavioural and toxicological effects, although slightly less potent, as cocaine and human subjects cannot discriminate between cocaine and cocaethylene given either by insufflation or intravenous injection (Jatlow *et al*. 1991; Landry 1992; Jatlow *et al*. 1996; Cami *et al*. 1998).

Treatment of cocaine and alcohol co-dependence

Treatment of a co-morbid disorder has yielded promising results in the treatment of substance dependence disorders (Boyarsky & McCance-Katz 2000). Recognition of concurrent cocaine and alcohol dependence is therefore important as it points to coordinated treatment of both dependencies. While behavioural therapies have effectiveness in the treatment of cocaine and alcohol dependence, specific medications with established effectiveness do not exist against cocaine dependence, but they do exist against alcohol dependence. Promising results have been obtained with disulfiram and naltrexone indicating that treating alcohol dependence will result in reduced cocaine use in dually addicted patients (see the review by Boyarsky & McCance-Katz 2000). For example, treatment with disulfiram appeared to have significant benefits, in terms of better retention in treatment, longer periods of abstinence from cocaine and alcohol, and reduced frequency and intensity of alcohol use over time (Carroll *et al*. 1998). Cocaine and alcohol use appeared to be associated strongly throughout treatment. Another preliminary study with 15 out-patient subjects showed a significant reduction in both alcohol and cocaine use after naltrexone treatment (Oslin *et al*. 1999). Efficacy could not be determined from that study because of lack of a placebo control group.

As the mesolimbic dopamine system is believed to have a key role in the mediation of the positive reinforcing stimulus effects of drugs with dependence potential, including alcohol and cocaine, it can be expected that pharmacological blockade of dopaminergic neurotransmission with dopamine D1 and/or D2 receptor antagonists may reduce drug intake and affect drug dependence. Flupenthixol has mixed dopamine D1/D2 receptor antagonist properties. The few studies so far suggest it has effectiveness as a consequence of a flupenthixol-induced extreme akathisia when smoking crack (Soyka & De Vry 2000).

Retrospective studies

Extent of use

A 1990 survey of the general population estimated that approximately 4 million US citizens had engaged in simultaneous use of alcohol and cocaine during the previous month, with 9 million having done so in the previous year. It was also estimated that 5 million had engaged in concurrent use during the previous month, with 12 million having done so in the past year (Grant & Harford 1990). In the more recent 1997 National Household Survey on Drug Abuse of more than 25 000 people, those who had used alcohol during the last month were six times as likely to have used cocaine during the last month as those who had not used alcohol. Those who admitted to heavy alcohol use during the past month were eight times as likely to have used cocaine during the last month as those who did not admit to heavy alcohol use (Substance Abuse and Mental Health Services Administration 1999).

The other set of data focuses on individuals who needed medical care or drug abuse treatment (Walsh *et al*. 1991; Higgins *et al*. 1994; Wiseman & McMillan 1996; Erickson *et al*. 1998; Heil, Badger & Higgins 2001). An analysis of 3762 emergency room patients in the United States between 1983 and 1993 found that 43% had used cocaine. Approximately one-third of these patients had used cocaine alone, another third cocaine plus alcohol, and the remaining third cocaine plus some other drug (usually a sedative such as diazepam) (Erickson *et al*. 1998). One study of alcoholics seeking treatment found that 40% of them had used cocaine during the last year (Walsh *et al*. 1991), meaning that such concurrent use of cocaine by alcoholics was seven or eight times greater than that shown by the general population (Grant & Harford 1990). An even higher percentage of cocaine abusers also use (and significantly, abuse) alcohol. That is, 88% of cocaine abusers in an in-patient rehabilitation programme reported that they engaged in co-use of alcohol (Wiseman & McMillan 1996). Furthermore, analyses of consecutive admissions for cocaine treatment revealed that 57% (Higgins *et al*. 1994) and 61% (Heil, Badger & Higgins 2001) of cocaine-dependent users were also alcohol-dependent. Also, the rate of simultaneous use of alcohol and cocaine by alcoholics was reported to be more than twice that of non-alcoholics, 87% versus 34% (Heil, Badger & Higgins 2001). The general view that alcohol use can increase cocaine use seems also to be applicable to the reverse, that is that cocaine use can increase alcohol use. This view is supported by the fact that about 60% of cocaine/crack-using patients in a methadone maintenance programme report using alcohol to come down from coke/crack, thus to temper the discomfort associated with tapering and ceasing cocaine/crack use (Magura & Rosenblum 2000).

Operating motor vehicles

Consistent with prospective research demonstrating cocaine's antagonist effect on alcohol-induced behavioural deficits, retrospective research examining the effect of co-use of alcohol and cocaine shows the substantial impact of alcohol alone (Cimbura *et al*. 1982; Mason & McBay 1984; Mercer & Jeffery 1995; Carmen & Alvarez 2000). However, the combination provides no definitive protection from the impairing effects of alcohol, as 21 of 600 fatally injured drivers in the US county of Los Angeles (Budd, Muto & Wong 1989) and 12 of 285 fatally injured drivers in Spain (Carmen & Alvarez 2000) tested positive for both alcohol and cocaine.

A 1988 study of urine samples from volunteer truck drivers showed evidence of cocaine use in 2% of the subjects, alcohol use in less than 1%, and no evidence of co-use (Lund *et al*. 1988). There was no evidence of cocaine alone or alcohol/cocaine combinations in military aircraft fatalities from 1986 to 1990 (Klette *et al*. 1992). Finally, a 1991 study of US train employees involved in 175 accidents revealed that 6.7% tested positive for 'drugs or alcohol', although no data were provided regarding the co-use of alcohol and cocaine (Moody *et al*. 1991).

Cocaethylene and toxicity

Analysis of 451 urine samples, obtained from patients admitted to a variety of locations in a university hospital and found positive for cocaine or its metabolite, revealed that 57 (12.6%) contained cocaethylene (Bailey 1995). Of these 57 cocaethylene positive samples, 53 samples were from emergency or trauma unit patients. The mean blood alcohol concentration in 37 cocaethylene-positive samples appeared to be significantly higher (1.3 g/l) than that in 117 cocaethylene-negative samples (0.6 g/l). In another study of 18 emergency room patients with positive blood alcohol and urine cocaine, 11 had cocaethylene in their plasma, two cocaine only and five neither cocaine nor coaethylene (Bailey 1993). Except for one subject, there was a good correlation between the concentration of plasma cocaine and cocaethylene. Consistent with this, analysis of over 1000 toxicological screens for emergency room patients revealed that 88% of the people who tested positive for alcohol and cocaine had produced cocaethylene. Furthermore, plasma cocaethylene levels correlated well with plasma cocaine levels (Signs *et al*. 1996). Inexplicably, a different laboratory reported no correlation between plasma cocaethylene and cocaine, even when subjects were stratified by alcohol levels (Brookoff *et al*. 1996).

There is some controversy as to whether cocaethylene, or alcohol-induced increases in cocaine levels, is responsible for the increased heart rate, and presumed increased cardiotoxicity, arising from the alcohol/cocaine combinations (Jatlow *et al*. 1991; Landry 1992; Jatlow *et al*. 1996; Cami *et al*. 1998). The latter part of the discussion may be premature, in that no retrospective studies demonstrate unequivocally that the combination in fact leads to greater cardiotoxicity. For example, a study involving over 1000 consecutive toxicological screenings of emergency room patients suggested a predominant role for cocaine alone. This study did not reveal more cardiovascular problems in subjects positive for both drugs than in subjects positive for one of the drugs (Signs *et al*. 1996). In another study, subjects who used the combination had less chest pain, but more back and extremity pain, than those who had used cocaine only (Vanek *et al*. 1996). The results of this last study are difficult to interpret, since there was no alcohol-only group and it is thus impossible to determine if this 'combination' effect arose from alcohol only.

There has been one recent report that an 18-year-old male who consumed cocaine, alcohol and marijuana died a sudden death from fatal cardiotoxicity (Daisley *et al*. 1998). This single report with one subject employing an unknown dose and administration schedule must be considered in the context of the previously mentioned study of 1000 patients in which a predominant role was found for cocaine alone in acute cardiotoxicity.

Violence

There is much theorizing about the possible mechanism by which the alcohol/cocaine combination leads to more violence than either drug alone, but this is sometimes carried out without an attempt to provide data from humans to support the initial hypothesis. For example, one theoretical paper which failed to present any human data regarding alcohol/cocaine interactions discussed none the less the role of 'kindling of a limbic dyscontrol system' and concluded that 'despite uncertainty about the mechanisms of alcohol-related enhancement of aggression, it is arguable that drinking may increase the probability of a cocaine-associated episode of violence via the disinhibitory action of alcohol' (Davis 1996, p. 1290). However, alcohol and cocaine each elevate extraneuronal dopamine and serotonin levels, and this may lead to deficits in impulse control and, thus, violent behaviour (Ritz, Kuhar & George 1992; Wozniak & Linnoila 1992).

Fortunately, retrospective studies have looked at human drug users to investigate the effect of alcohol/cocaine combinations on violence. Some of these studies determined that there was a predominant role for alcohol, while cocaine added only its own effects, without evidence of a potentiating interaction (Abel 1987; Rodenas, Osuna & Luna 1989; Chambers 1990; Bradford, Greenberg & Motayne 1992; Yarvis 1994; Yu 1998; ElSohly & Salamone 1999; Heil, Badger & Higgins 2001). One study of 650 homicide victims found that 34.6% had used alcohol alone, 31.3% cocaine alone, and 10.6% the combination (Tardiff *et al*. 1995). A study of suspects in 20 lethal domestic violence cases found that nine suspects (56%) had used alcohol alone, one (6%) cocaine alone, five (31%) the combination and one (6%) neither alcohol nor cocaine (Slade, Daniel & Heisler 1991). However, the results of these two studies are difficult to interpret because 'there is a lack of comparative knowledge of base drinking (and cocaine using) rates in these populations represented in crime statistics' (Bradford, Greenberg & Motayne 1992, p. 610).

Two studies suggest that there is a potentiating effect of the alcohol/cocaine combination upon violent behaviour, but are difficult to interpret due to lack of appropriate control groups. The first of these found that for patients presenting to an emergency room with trauma, 22% had used cocaine only and 50% had used the alcohol/cocaine combination. When violent trauma was considered, 17% had used cocaine only and 37% the combination (Vanek *et al*. 1996). However, the effect of the combination may have been due to alcohol alone. Similarly, a study of 320 male veterans in treatment for cocaine abuse found that those who used the combination rather than cocaine alone were more likely (40% versus 26%) to beat someone severely and more likely (39% versus 24%) to threaten someone with a weapon (Denison, Paredes & Booth 1997). The authors stated that 'use of alcohol seems to increase the likelihood of the cocaine user engaging in violent behavior' (p. 301). Again, no attempt was made to determine if the effect of the combination may have been due to alcohol alone. Therefore, the violence attributed to cocaine may in part be a product of alcohol abuse and alcoholism (Heil, Badger & Higgins 2001).

One well-designed study determined that the alcohol/cocaine combination had a potentiating effect upon violent thoughts. Despite no differences in suicidal ideation, subjects who used the combination were three times as likely as those who took alcohol only, and five times as likely as those who took cocaine only, to have homicidal ideas or plans (Salloum *et al*. 1996). This demonstration of a potentiating effect of the combination upon homicidal ideation must be replicated, and an attempt must be made to determine if the influence of combined use on thoughts goes on to influence behaviours. Interestingly, in a study of 41 depressed alcoholics, it was found that cocaine use made it twice as likely that these alcohol abusers would attempt suicide, although no data was presented on the role of alcohol in suicide attempts by cocaine abusers (Cornelius *et al*. 1998).

Neuropsychology and psychiatry

One study indicated that users of alcohol/cocaine combinations do not exhibit more psychiatric problems than those who use one compound only (Salloum *et al*. 1996). Others found that alcohol-dependent cocaine users did show more severe psychological problems than non-alcoholic cocaine users (Carroll, Rounsaville & Bryant 1993; Heil, Badger & Higgins 2001). In a consecutive series of 214 cocaine dependent patients admitted to a substance abuse treatment programme, 58% of the suicide attempters had a life-time history of alcohol dependence compared to 35% of those who had never attempted suicide (Roy 2001). Another study showed additive, dose-related detrimental effects of cocaine and alcohol on neurobehavioural performance after 1–3 days of abstinence, with these effects persisting after 4 weeks of abstinence (Bolla, Funderburk & Cadet 2000). A study which was concerned primarily with liver function showed that those who used the combination displayed more irritability and greater memory loss than those who used cocaine alone, but this study did not include an alcohol-only comparison group (Tabasco-Minguillan, Novick & Kreek 1990).

A recent study of 335 incarcerated male felons found that subjects who co-used alcohol and cocaine performed significantly worse upon tests of short-term memory, long-term memory and visual motor ability than subjects who used only one of the compounds (Selby & Azrin 1998). In contrast, a study of abstinent crack-dependent versus abstinent alcohol–crack-dependent subjects found no significant differences in performance upon a battery of neuropsychological tests (Di Sclafani *et al*. 1998).

There are four studies that reveal the apparently paradoxical finding that users of the combination may in some ways be better off than those who use one compound only. For example, one study of 64 subjects in a drug abuse rehabilitation centre found that the patients who had used the combination showed fewer neuropsychological deficits than did those who used alcohol only or cocaine only (Easton & Bauer 1997). Another study showed poorer neuropsychological performance among singly addicted cocaine users compared to co-dependent cocaine and alcohol users (Robinson, Heaton & O'Malley 1999). Similarly, 124 consecutive admissions for cocaine treatment showed that there were fewer seizures in those who combined alcohol and cocaine most frequently (Higgins *et al*. 1994). However, this latter observation has not been replicated and may have been due to sample size (Heil, Badger & Higgins 2001). The latter study actually suggests a higher seizure probability among those who combine alcohol and cocaine more frequently. The fourth study looked at 55 males withdrawing from alcohol alone versus 94 withdrawing from alcohol and cocaine, and found that withdrawal was less severe in those who had combined the two compounds (Castaneda *et al*. 1995). Some of these results may reflect the antagonist effects of each of the compounds upon some effects of the other, as for example alcohol's vasodilatory effect and cocaine's vasoconstrictive effect. Another explanation may be that the superior performance of users of the combination may arise from confounding differences in drug administration and dosage. For example, users of the combination might consume less alcohol over their drug-abusing career, and thus experience less severe alcohol withdrawal (Castaneda *et al*. 1995). Also, the authors of the study who found superior neuropsychological test performance in users of the combination suggested that these individuals were less likely than those who used cocaine alone to smoke 'crack', and thus may have been exposed to lower dosages and lower peak brain levels of cocaine than their counterparts (Easton & Bauer 1997).

Finally, a study of 98 males in intensive out-patient rehabilitation programmes for cocaine abuse found that the consumption of alcohol had no effect upon relapse in cocaine abusers who were not previously alcohol-dependent. On the other hand, past cocaine abusers who had been alcohol-dependent were 6–10 times more likely to report that alcohol consumption occurred prior to a relapse into cocaine abuse (McKay *et al*. 1999).

Liver function, HIV risk and teratology

The only study to examine the effect of non-parenteral alcohol/cocaine use upon liver function found that the combination was not more likely to induce liver pathologies than either of the compounds alone (Tabasco-Minguillan, Novick & Kreek 1990).

A recent study among 495 African-American crack abusers showed that those reporting frequent use of both alcohol and cocaine had higher rates of HIV risk-related sexual behaviours than those reporting frequent use of alcohol only or crack only. That is, frequent co-users were more likely than the frequent alcohol users (with less frequent crack use) or the frequent crack users (with less frequent alcohol use) to report alcohol- and crack-impaired sex, two or more sexual partners and greater number of sexual acts without condoms (Rasch *et al*. 2000).

One report presented a single case of a child born to a mother who co-used cocaine and alcohol throughout her pregnancy. This male child was born at 32 weeks and at 4 months had a variety of facial abnormalities, some of which could be attributed to the fetal alcohol syndrome, some to the known teratogenic effects of cocaine, and some of which were unique features that may be attributable to co-use (Robin & Zackai 1994). Further research is needed to confirm this supposedly potentiating teratogenic effect of co-use of alcohol and cocaine.

CONCLUSIONS

Survey research in normal populations reveals a high percentage of concurrent and simultaneous use of alcohol and cocaine. Patients seeking treatment for abuse of alcohol are likely to have used cocaine during the past year, while patients seeking treatment for cocaine abuse are often also alcohol-dependent. Patients who were once dependent upon both cocaine and alcohol, compared to those previously dependent upon cocaine alone, reveal a greater tendency for alcohol use to trigger a relapse of cocaine abuse.

In general, prospective experiments show that individuals who combine alcohol and cocaine perceive a more intense feeling of intoxication, that cocaine has either no effect or an antagonist effect on alcohol inebriation and that cocaine antagonizes the deficits in learning and psychomotor performance induced by laboratory doses of alcohol. Consistent with these experimental findings, retrospective research suggests a major role for alcohol alone in driving deficits without evidence that co-use of cocaine adds to alcohol-induced driving deficits.

Prospective research shows that simultaneous use of cocaine and alcohol leads to heart rate increases that are statistically additive, yet show a trend towards greater than additive. These heart rate effects are not seen when cocaine is given 30 minutes before alcohol. One study revealed an additive effect of the combination on heart oxygen demand, but it also showed a concomitant increase in epicardial coronary arterial diameter.

Simultaneous use (cocaine by snorting) leads to elevated blood levels of cocaine, but not of alcohol. Intranasal cocaine 30 minutes before alcohol does not give elevated blood cocaine levels compared to cocaine alone. Cocaethylene is formed by subjects given both alcohol and cocaine. Emergency room and post mortem studies reveal that cocaethylene is usually, but not always, found in patients testing positive for both alcohol and cocaine.

Retrospective studies suggest that combined use does not cause more cardiovascular problems than expected from the additive effects of each drug. Data suggesting that the combination causes greater health problems for the patient in the emergency room are flawed by a lack of proper controls for the effects of alcohol alone.

Retrospective research on violence suggests a major role for alcohol alone and additive increases in violent behaviour or crime from the combination. For ethical reasons, high repeated doses of cocaine have not been studied experimentally. The behavioural toxicity of high-dose binge use of cocaine with alcohol may thus be underestimated. One study shows a potentiation effect of the cocaine-alcohol combination on the number of thoughts and threats of violence.

One retrospective study suggests that combined non-parenteral use is no more likely than either drug alone to induce liver damage or neuropsychological deficits. One other report suggests that the teratogenic effects of combined use are more than additive with unique features not seen with alcohol alone or cocaine alone.

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